

Replication Plan

Latent Stochastic Differential Equations for Modeling Quasar Variability and Inferring
Black Hole Properties

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Auto-generated Replication Blueprint

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Contents

1	Paper Overview	2
2	Detailed Workflow	2
2.1	Phase 1: Stochastic Process Models	2
2.2	Phase 2: Simulated Light Curve Generation	2
2.3	Phase 3: Parameter Inference	2
2.4	Phase 4: Application to Real Data	2
2.5	Phase 5: Validation	3
3	Datasets	3
4	Models, Tools, and Software	3
4.1	Hardware Requirements	4

Paper Overview

This paper develops latent stochastic differential equation (SDE) models for quasar optical variability and uses them to infer physical properties of supermassive black holes (mass, accretion rate). Quasar light curves are modeled as observations of a latent SDE process, going beyond the standard damped random walk (DRW/OU process) to capture more complex variability. The model is fit using likelihood-based inference (possibly with the **celerite** fast GP framework for scalable computation). The authors validate on simulated light curves and apply to real survey data (e.g., SDSS Stripe 82, ZTF).

Detailed Workflow

Phase 1: Stochastic Process Models

1.1 Implement the DRW (OU process) baseline.

- $dX(t) = -\frac{1}{\tau}X(t)dt + \sigma dW(t)$
- Analytic covariance function: $C(\Delta t) = \hat{\sigma}^2 e^{-|\Delta t|/\tau}$.

1.2 Implement higher-order latent SDEs.

- CARMA(p, q) models or custom multi-dimensional SDEs.
- These have richer power spectral densities (PSDs).

1.3 Implement the **celerite** representation for fast ($O(N)$) likelihood evaluation.

Phase 2: Simulated Light Curve Generation

2.1 Generate mock light curves from each SDE model.

- Sample at irregular time cadences mimicking survey strategies.
- Add photometric noise with realistic magnitude-dependent errors.

2.2 Inject known physical parameters (M_{BH} , $\dot{M}$, τ , σ).

Phase 3: Parameter Inference

3.1 Likelihood computation using **celerite** or direct GP.

3.2 Parameter optimization (maximum likelihood) and MCMC sampling (posterior distributions).

- Use **emcee** or **PyMC** for MCMC.
- Sample SDE parameters and physical parameter mappings.

3.3 Model comparison (DRW vs. higher-order SDE) via BIC, AIC, or Bayes factors.

Phase 4: Application to Real Data

4.1 Download quasar light curves from SDSS Stripe 82 or ZTF.

4.2 Fit models to real light curves.

4.3 Infer black hole masses and compare to spectroscopic estimates.

4.4 Population-level analysis of inferred parameters.

Phase 5: Validation

5.1 Parameter recovery tests on simulated data.

5.2 Reproduce paper’s figures (posterior distributions, PSD comparisons, mass correlations).

Datasets

Dataset / Source	Type	Location / Notes
Simulated quasar light curves	Synthetic	Generated from SDE models with known parameters
SDSS Stripe 82 light curves	Public survey	https://www.sdss.org/dr17/Quasar catalog : https://faculty.washington.edu/ivezic/macleod/qso_dr7/
ZTF light curves	Public survey	https://www.ztf.caltech.edu/
Quasar catalog (spectroscopic)	Public	SDSS DR16Q: https://www.sdss.org/dr16/algorithms/qso_catalog/
Black hole mass estimates	Published	Shen et al. (2011) SDSS quasar catalog

Models, Tools, and Software

Tool / Library	Version	Purpose / URL
celerite / celerite2	latest	Fast GP/SDE likelihood computation https://celerite2.readthedocs.io/
emcee	≥ 3.1	MCMC sampler https://emcee.readthedocs.io/
PyMC	≥ 5.0	Bayesian modeling framework (alternative) https://www.pymc.io/
JAX	latest	Autodiff and JIT compilation for likelihood https://github.com/google/jax
NumPy / SciPy	≥ 1.21	Numerical operations
Matplotlib	≥ 3.4	Light curve and posterior plots
Astropy	≥ 5.0	Astronomical data I/O and coordinates
corner.py	latest	Corner plots for MCMC posteriors https://corner.readthedocs.io/

Hardware Requirements

- MCMC on individual light curves: single CPU core, minutes per object.
- Population analysis ($\sim 10,000$ quasars): multi-core workstation or cluster.